Effect of private information on indirect reciprocity

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Indirect reciprocity is one of the key mechanisms for the evolution of cooperation. It relies on mutual monitoring and assessments, i.e., individuals collect information about the past behavior of others and judge whether that behavior is "good" or "bad." A player will not be helped if labeled with a bad image. There are many ways for assessing others, each of which can be interpreted as an elementary form of a moral sense (i.e., a view on what is good or bad). The information can be either public or private: private information can lead to mismatches between the opinions of individuals even when they share the same moral sense. In this paper, the effect of private information on the best-known assessment rules is investigated. In order to calculate payoffs, the concept of an image matrix is introduced. It describes who is good in the eyes of whom, and its time evolution is given by a probabilistic Boolean automaton. In contrast to the public information case, private information leads to the collapse of the sterner assessment rule. Alternatively, stable polymorphisms may subsist, with the milder rule and a more simple-minded rule coexisting together with unconditional cooperators; thus, cooperation can be sustained by indirect reciprocity even in the absence of public information.

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I. INTRODUCTION

Indirect reciprocity is known as one of the fundamental mechanisms that sustain mutual cooperation [1–4]. One feature of indirect reciprocity is that helpful acts are returned, not by the recipient as in direct reciprocation, but by third parties: if X helps Y, then X is helped in turn by some Z, rather than by Y. The impact of indirect reciprocity has been extensively discussed in the literature, both for human populations [5–9] and for other species [10]. Indirect reciprocity is based on the assumptions that individuals in a population pay attention to the interactions among others, and that they assess those interactions even when they are not personally engaged in them [11–14]. If the action of an individual is judged as bad by others or, in other words, if that individual is labeled with a bad image, it will not be helped.

Thus, in indirect reciprocity, players must collect information about the behavior of others. Using this information, they determine their actions. To do this requires the following two "modules:" (a) an assessment rule by which to judge acts of others as good or bad, even when they are not directed at oneself, and (b) an action rule specifying how to act toward others, based on that assessment. This yields a rudimentary model of a moral sense. Clearly there are many possibilities for those rules [15,16]. Nowak and Sigmund [12] investigated the simplest assessment rule: it only takes into consideration whether help is given or not. In the first case, the donor is regarded as good and in the second case as bad. The action rule prescribes to help only good individuals. However, this strategy (i.e., the combination of assessment and action rule) displays an intriguing inconsistency since refusing to help is always perceived as bad. Therefore, when Z refuses to help X because X is a bad person, Z is perceived by third parties as bad, although they would have acted in the same way.

This fact suggests us to study more sophisticated assessment rules which can distinguish between a justified refusal and an unjustified refusal. Such assessment rules deal with higher-order information. In the literature, assessment rules are classified into three types, depending on what information they use [15]. A first-order assessment rule only takes into account whether X helps Y or not (as in the above example). A second-order assessment rule takes also into account the image of the recipient Y [3,17]. A third-order assessment rule in addition takes into account the image of the donor X [11,18].

Assuming binary assessments, i.e., a world in black and white, Ohtsuki and Iwasa [16,18] showed that among the resulting 4096 possible strategies, only eight lead to a stable regime of mutual cooperation. These are called the leading eight. Two of these strategies are based on second-order assessment, none on first-order assessment. In order to discuss the advantages and disadvantages of these leading eight strategies, Uchida and Sigmund [19] proposed an analytical approach to compare different assessment rules. They applied this method to the competition of the two second-order assessment rules as well as to the above-mentioned first-order assessment rule. They verified that the first-order assessment rule does not lead to a stable cooperative state. Comparing the two second-order rules, Uchida and Sigmund concluded that the sterner second-order rule wins: players using the sterner rule are more likely to earn a higher payoff than players following the other strategy. Other papers based on numerical simulations also argue the merit of the sterner rule for group structured populations [15,20,21].

All these studies assume that the information is perfect and public: all members in a population know the previous action of all others. As a result they share exactly the same information and determine their actions by using this public information. However, daily experience tells us that persons can have different information, and that such differences lead to a mismatch between their opinions even when they share the same "moral sense." What happens if their information is

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TABLE I. Definitions of the five assessment rules studied in this paper.

Situation	All C	All D	SCORING	TYPE I	TYPE II
Give help to good recipient	Good	Bad	Good	Good	Good
Give help to bad recipient	Good	Bad	Good	Good	Bad
Refuse help to good recipient	Good	Bad	Bad	Bad	Bad
Refuse help to bad recipient	Good	Bad	Bad	Good	Good

not public but only private? Do the merits of the leading eight strategies, especially of the sterner rule, also hold in those cases? Is it still impossible for the first-order assessment to subsist? These are the questions addressed in this paper. Assuming that each individual observes different parts of the past series of interactions, I compare the two second-order assessment rules in the leading eight as well as the first-order assessment rule, both by numerical simulations and by an analytical approach for a limit case. In order to calculate payoffs, the concept of an image matrix is introduced, which describes who is good in the eyes of whom. Its time evolution is given by a probabilistic Boolean automaton. It will be shown that, in contrast to the case of public information, the sterner rule now loses its advantage and does not lead to stable cooperation. In the absence of public information, a high cooperation level can be stably sustained by a stable coexistence of the other second-order assessment rule with the first-order assessment rule and with unconditional cooperators. The following sections describe the model and the methods to analyze it, then derive the results, and discuss both the model and outcomes.

II. MODEL

As done by Uchida and Sigmund [19], a large well-mixed population is considered. From time to time, two individuals meet randomly and engage in a one-shot donation game: a coin toss decides who plays the role of (potential) donor and who is the recipient. The donor chooses whether or not to help the recipient at a personal cost c. A helpful action provides the recipient with a benefit $b>c$. Each individual experiences such interactions many times, always with different partners.

Indirect reciprocity presumes that players are able to observe each other and evaluate each other's action. Each player Z has an assessment rule by which to judge others, using their behavior in the role of the donor. Player Z's judgment is binary: it assigns the label of either good or bad to other players. The players can have different assessment rules. In this paper, the following assessment rules will be considered (see Table I):

(1) All C (regard everyone as good);

(2) All D (regard everyone as bad);

(3) SCORING (regard exactly those as good who, in their previous round, gave help);

(4) TYPE I, known as simple standing or SUGDEN (regard everyone as good except those who, in their previous round, refused help to a good recipient) [3]; and

(5) TYPE II, also called KANDORI (view exactly those as good who, in their previous round, gave help to a good recipient or refused help to a bad recipient) [17,18].

Note that TYPE II is sterner than TYPE I. They only differ in their treatment of those who give help to a bad player: TYPE I appreciates this, but TYPE II condemns it.

In addition, players have action modules that determine how they act as donors. Our strategies all use the action rule to help if and only if the recipient is good. TYPE I and TYPE II are the only second-order assessment rules belonging to the leading eight strategies: these are the strategies with the property that, in a homogeneous population, it does not pay to use another action module [16].

As usual, players are assumed to sometimes commit an implementation error: with a certain probability ϵ, they fail to implement an intended help. Following Leimar and Hammerstein [11], Ohtsuki and Iwasa [16,18], and Panchanathan and Boyd [14], an intended refusal is always carried out (see also Fishman *et al.* [22,23] and Lotem *et al.* [24]). In this paper, ϵ is set to 0.1 for simulations and numerical analysis.

Each interaction will only be observed by a fraction q of the population. This assumption is different from that of Uchida and Sigmund [19] (who assumed that all individuals perfectly observe all interactions) and that of Ohtsuki and Iwasa [16,18] (who assumed that one player acts as a referee whose assessment is perfectly copied by all other players). In both approaches, all the members in the population thus share the same information. Clearly, $q=1$ corresponds to the case of perfect information. In this paper, the effect of imperfect information, i.e., $q<1$ will be investigated.

To specify the payoffs, let r_{ij} $(i,j=1,2,\ldots,5)$ denote the probabilities that a randomly chosen individual using strategy i positively assesses another randomly chosen player using strategy j. The payoff P_i is then given by

$$P_i = -\bar{\epsilon}\sum_{j=1}^{5} x_j r_{ij} c + \bar{\epsilon}\sum_{j=1}^{5} x_j r_{ji} b, \tag{1}$$

where x_i is the frequency of strategy i in the population and $\bar{\epsilon} := 1-\epsilon$ is the probability that an intended help is actually given. The first term $\bar{\epsilon}\Sigma_j x_j r_{ij}$ is the probability that a player of type i gives a help to another player, and thus incurs cost c. The second term $\bar{\epsilon}\Sigma_j x_j r_{ji}$ is the probability that a player of type i is helped by a randomly chosen donor, and thus provided with a benefit b.

Specifying the values of r_{ij} determines the payoffs. This allows us to study the time evolution of the state of the society $(x_1,x_2,\ldots,x_5)$ by the replicator dynamics $\dot{x}_i = x_i(P_i - \bar{P})$, where $\bar{P}=\Sigma_k x_k P_k$ is the average payoff in the popula-

tion. This dynamics describes cultural evolution (or social learning by imitation): successful strategies spread [25,26].

In general, r_{ij} also changes with time due to the observation process, but we shall assume that such changes are much faster than the changes in the frequencies of strategies x_i caused by the above-mentioned cultural evolution. Thus, for a given x_i, r_{ij}'s are assumed to have reached equilibrium after many interactions. The next section presents numerical simulations to determine the values of r_{ij}, and for the limiting case of small q an analytical approach.

III. METHODS

By the definitions of All C and All D, r_{ij} is 1 and 0 for $i=1$ and $i=2$ respectively, irrespective of j. For the discriminating strategies SCORING, TYPE I, and TYPE II, i.e., for $i=3,4,5$, r_{ij}'s can be determined by numerical simulations. In the limiting case of small q, an analytical calculation is feasible.

A. Boolean automaton: Numerical simulations

Let β_{XY} denote the image of player Y in the eyes of X. The matrix (β_{XY}) will be called the image matrix. It determines r_{ij}. The time evolution of the image matrix is described by the following probabilistic Boolean automaton: let u, $v \neq u$, and $\mathcal{W}$ denote the donor, the recipient, and the set of observers, respectively, chosen at random at time τ. The number of observers is qN, where N is the total number of individuals in the population. Then the image matrix at time $\tau+1$ is given by

$$\beta_{ZX}(\tau+1) = \begin{cases} \beta_{ZX}(\tau) & (Z \notin \mathcal{W} \vee X \neq u) \\ A^{(Z)}(\alpha_u, \beta_{Zv}) & (\text{otherwise}), \end{cases} \tag{2}$$

where α_u is the action of u toward v. Using the abbreviation C for "help," D for "refuse," 1 for good, and 0 for bad, $\alpha_u = D$ always if $\beta_{uv}=0$, or with probability ϵ if $\beta_{uv}=1$. Otherwise, $\alpha_u = C$. In Eq. (2), the function $A^{(Z)}(\alpha,\beta) \in \{0,1\}$ with $\alpha \in \{C,D\}$ and $\beta \in \{0,1\}$ is the assessment, from the viewpoint of Z, if the donor chooses action α toward the β recipient. For instance, if Z uses TYPE II, $A^{(Z)}(C,1) = A^{(Z)}(D,0) = 1$ and $A^{(Z)}(C,0) = A^{(Z)}(D,1) = 0$ (see Table I).

Note that for any given updating step, only the u column of the image matrix changes its values. Since the action α_u is probabilistically determined by donor u's assessment of recipient v, the new assessment $\beta_{wu}(\tau+1)$ in the eyes of observer $w \in \mathcal{W}$ (probabilistically) depends on the assessment of recipient v both in the eyes of u and w before the game. For example, for a TYPE II observer w, if $\beta_{uv}=1$ and $\beta_{wv}=0$, the new assessment $\beta_{w,u}(\tau+1)$ is 0 with probability $\bar{\epsilon}$ and 1 with probability ϵ [Fig. 1(c)]. Figure 1 illustrates how the image matrix is updated by Eq. (2) for all cases. This yields a stochastic process on the image matrix which determines who has a good image in the eyes of whom. This process describes the assessment dynamics.

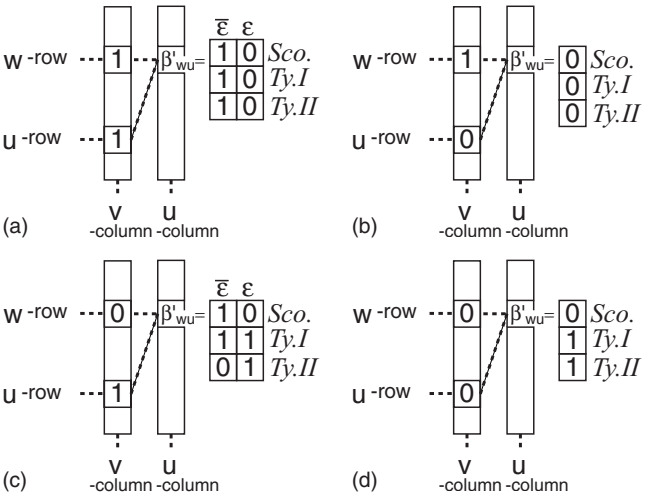


FIG. 1. Update of the image matrix. The donor, the recipient, and one of the observers chosen at time τ are denoted by u, v, and w respectively. In the figure, β'_{wu} denotes the new assessment of donor u in the eyes of observer $w \in \mathcal{W}$, and Sco. means that w uses SCORING, Ty.I means TYPE I, and Ty.II denotes TYPE II. The new assessment $\beta'_{wu} = \beta_{wu}(\tau+1)$ depends on the current assessment of v both in the eyes of $u(\beta_{uv})$ and $w(\beta_{wv})$: (a) $\beta_{uv}=\beta_{wv}=1$, (b) $\beta_{uv}=0, \beta_{wv}=1$, (c) $\beta_{uv}=1, \beta_{wv}=0$, and (d) $\beta_{uv}=\beta_{wv}=0$.

In the computer simulations, the above stochastic process is run with the frequencies $(x_1,\ldots,x_5)$ kept fixed. In Figs. 5 and 6, the total population size is set to 1000, and each individual observes $T=10^5$ games on average. That is, in the case of $q=m/n$ $(n \geq m)$, the total number of iterations in a simulation is $n/m \times 10^5$. This number is not intended as a realistic one, but used so that the stochastic process reaches a stationary state: if T is modified, for example, to 2×10^5, the outcome of the simulations does not change. All individuals are assumed to be initially trustful, i.e., all the images β_{XY} are 1 initially. Due to the implementation error, the results do not depend on the initial condition.

B. Mean-field approach: Analytical calculation for small q

If only a very limited number of individuals observe each game (i.e., if q is small), it is unlikely that two individuals Z and Z' chosen from the large population observe X's action at the same time, i.e., they never share the same information. Since X's opponent Y will be selected at random at each time from the whole population and since X's action depends only on the image of Y, the probabilities that Z and Z' have good images of X will be independent, even when they follow the same assessment rule.

The following equality holds at equilibria:

$$r_{ij} = (\text{prob. that } j \text{ actually helps and } i \text{ positively assesses the action}) + (\text{prob. that } j \text{ defects and } i \text{ positively assesses the action}). \tag{3}$$

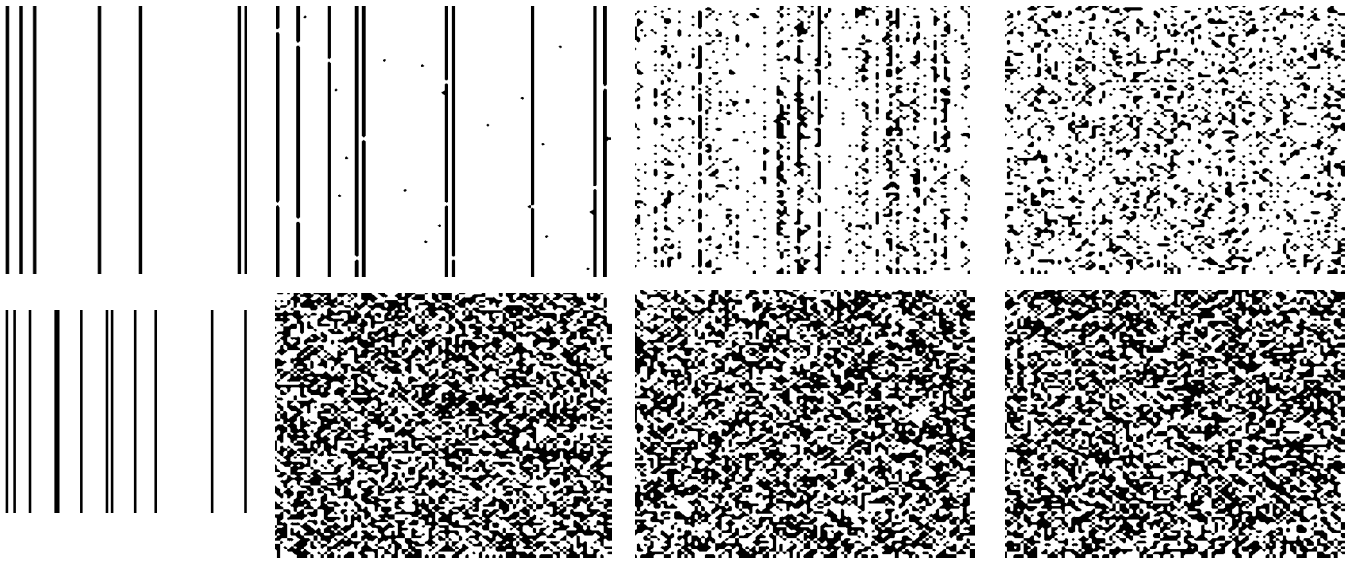


FIG. 2. The image matrix (β_{XY}) after 2×10^4 iterations. Top: TYPE I and bottom: TYPE II. The values of q are 1, 0.99, 0.5, and 0.1 from left to right, respectively. The good image $\beta_{XY}=1$ corresponds to a white dot and the bad image $\beta_{XY}=0$ corresponds to a black dot. Each population has 100 individuals.

Here, i and j in the right-hand side denote a randomly chosen player of types i and j, respectively. On account of the independence mentioned above, the first term is given by

$$\sum_k x_k \bar{\epsilon} r_{jk}[r_{ik}A^{(i)}(C,1)+(1-r_{ik})A^{(i)}(C,0)], \tag{4}$$

and the second term is given by

$$\sum_k x_k(1-\bar{\epsilon} r_{jk})[r_{ik}A^{(i)}(D,1)+(1-r_{ik})A^{(i)}(D,0)]. \tag{5}$$

For instance, in Eq. (4), x_k is the probability that j encounters k, $\bar{\epsilon} r_{jk}$ that j actually helps k, and r_{ik} that i has a good image of the potential recipient k. Inserting Eqs. (4) and (5) into Eq. (3), an equation system is obtained, which relates the unknowns r_{ij}'s. For example, for $i=4,j=3$ this yields $r_{43}=\bar{\epsilon}\Sigma_k x_k r_{3k}+\Sigma_k x_k(1-\bar{\epsilon} r_{3k})(1-r_{4k})$. The solution of these equations gives the payoff values. This approach is closely related to the annealed approximation method in random Boolean networks [27,28]. For pure populations (i.e., $x_3=1$, $x_4=1$, and $x_5=1$), the equations reduce to

$$r_{33}=\bar{\epsilon} r_{33}, \tag{6}$$

$$r_{44}=\bar{\epsilon} r_{44}+(1-\bar{\epsilon} r_{44})(1-r_{44}), \tag{7}$$

$$r_{55}=\bar{\epsilon} r_{55}^2+(1-\bar{\epsilon} r_{55})(1-r_{55}). \tag{8}$$

IV. RESULTS

A. Stationary image matrix: Homogeneous populations

Let us first explore the cases of homogeneous populations. In Fig. 2, some examples of the image matrix (β_{XY}) at the stationary state are shown, obtained by numerical simulations. The good image $\beta_{XY}=1$ is represented by a white dot in the corresponding place of the image matrix, and the bad image $\beta_{XY}=0$ is represented by a black dot.

The pure SCORING population always consists of bad players, independent of q (not shown in Fig. 2). For a population of TYPE I, the consensus in the columns of the matrix gradually disappears as q becomes smaller. On the other hand, for the TYPE II population, the image matrix changes dramatically if q becomes smaller than 1: consensus is completely destroyed and the black and white dots are randomly distributed.

The proportion of good players in the pure TYPE II population is always 1/2, for any $q<1$ and any ϵ (see also the supplementary material [29]). For the TYPE I population, it is a function of q and the error rate ϵ (Fig. 3). It is easy to see that in the limit case of small q, the proportion of good players is given by $1/(1+\sqrt{\epsilon})$. This is found by solving Eq.

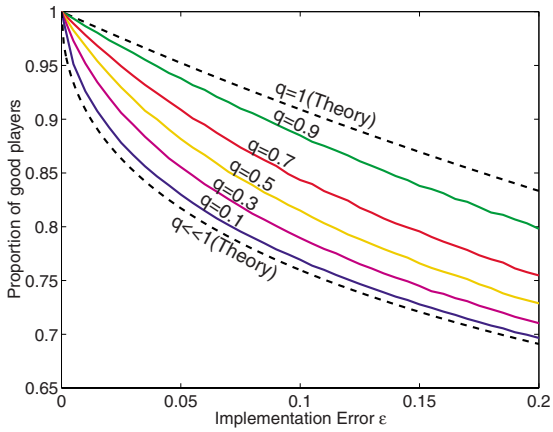


FIG. 3. (Color online) The proportion of good players in the pure population of TYPE I as functions of implementation error ϵ with different q's. The curves are obtained analytically for the perfect information case $q=1$ and the limit case $q\ll1$, for all other cases by simulations.

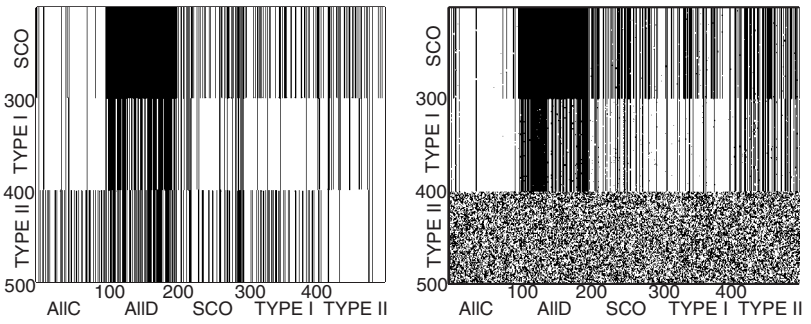


FIG. 4. The image matrix (β_{XY}) $(X=201,\ldots,500$ and $Y=1,\ldots,500)$ after 5×10^4 iterations. The left figure shows the image matrix for the case of perfect information $(q=1)$, and the right shows that for the case of imperfect information $(q=0.99)$. Each strategy is used by 100 individuals and there are 500 individuals in the whole population. The index of individuals Y runs from 1 to 500 (1–100: All C players, 101–200: All D players, etc.) and X runs from 201 to 500.

(7). For the case of perfect information, the proportion of good images is given by $1/(1+\epsilon)$ [18,30].

B. Stationary image matrix: Mixed populations

In Fig. 4, examples of the image matrix (β_{XY}) for mixed populations are shown for the perfect case $q=1$ and for the almost perfect case $q=0.99$. The matrix is divided into 15 blocks (or submatrices), each of which can be assigned an index (i,j), where $i=3,4,5$ and $j=1,\ldots,5$. Block (i,j) shows whether or not players of type i have good images of the players of type j. For example, the top-left block in Fig. 4 shows how SCORING players are assessing All C players, and the bottom-right block shows the assessments of TYPE II players in the eyes of TYPE II players. The white area of block (i,j) is proportional to r_{ij}. If the block (i,j) is totally white (or black), then r_{ij} is 1 (or 0).

For the perfect information case, synchronization is again realized in each block: all players of the same type have the same image of another individual. Note that, in this case, r_{ij} can be obtained analytically and that TYPE II individuals play better than TYPE I individuals (see Uchida and Sigmund [19]). In the case of imperfect information, the values of r_{ij} are obtained by numerical simulations. Even in the case of $q=0.99$, the consensus in the blocks of TYPE II is completely destroyed. But SCORING and TYPE I still show synchronizations in the corresponding blocks, respectively, with a small flaw due to the imperfectness of information. Moreover TYPE II players assess other players just at random regardless of which strategy they use. In fact it is numerically found that TYPE II has a good image with a probability of 1/2 of any player, i.e., TYPE II is a mixed strategy playing All C and All D with the equal weight. In order to show the time evolution of the image matrix for mixed populations, movies are provided as supplementary materials [29].

The difference between SCORING or TYPE I on one hand and TYPE II on the other is that, for the former strategies, a mismatch in the opinion between two observers can be repaired, while for the latter strategy this will not happen. Indeed, if two observers use SCORING or TYPE I assessment, and if they have different opinions toward some player, the difference will be removed whenever that player chooses action C. (In the case of SCORING, the mismatch will also be repaired if the player chooses D.) On the other hand, if two observers of TYPE II disagree about a given player, this difference remains, no matter which action the observed player chooses. Moreover, a new mismatch can be created between two individuals if an interaction is observed by one but not the other. Therefore, the mismatch between individuals spreads in the population of TYPE II until the image matrix becomes totally random. Thus, it is impossible for TYPE II to lead to stable cooperation. For this reason, the strategy TYPE II will henceforth be excluded from the numerical simulations.

C. Replicator dynamics

Vector fields generated by the replicator dynamics for $q=0.9$ and $q=0.1$ are shown in Figs. 5 and 6. Figures for $q=0.7$, 0.5, and 0.3 are presented as a supplementary material [29]. To produce the figures, simulations are performed 500 times at each point $(x_i=s_i\delta)$ in the state space, where $\delta=0.1$ and $s_i=0,1,\ldots,10$. For each run, the white area of block (i,j) is measured at the stationary state, and then averaged over the 500 simulations at each point in the state space to estimate r_{ij}.

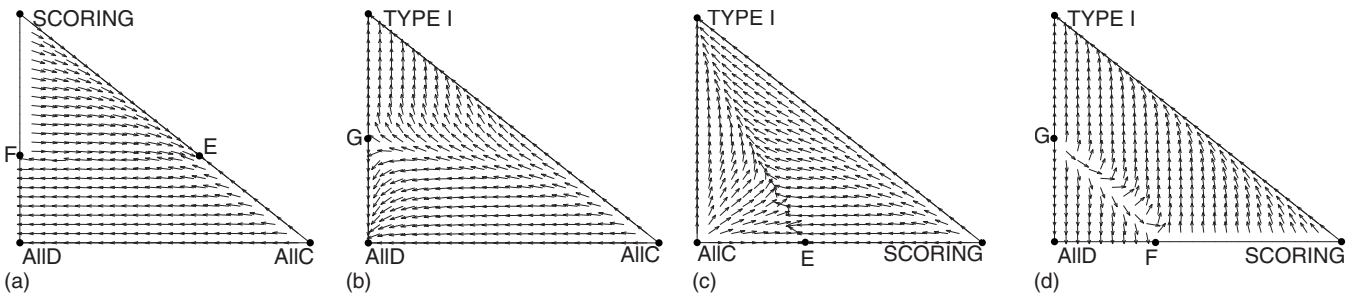


FIG. 5. The vector field defined by $x_i(P_i-\bar{P})$ on the surface: (a) All C-All D-SCORING, (b) All C-All D-TYPE I, (c) All C-SCORING-TYPE I, and (d) All D-SCORING-TYPE I. The vector is normalized at each point (except for the case where the vector vanishes), so that the direction is easily recognized. Parameters: $N=1000$, $q=0.9$, $b=3$, $c=1$, and $\epsilon=0.1$.

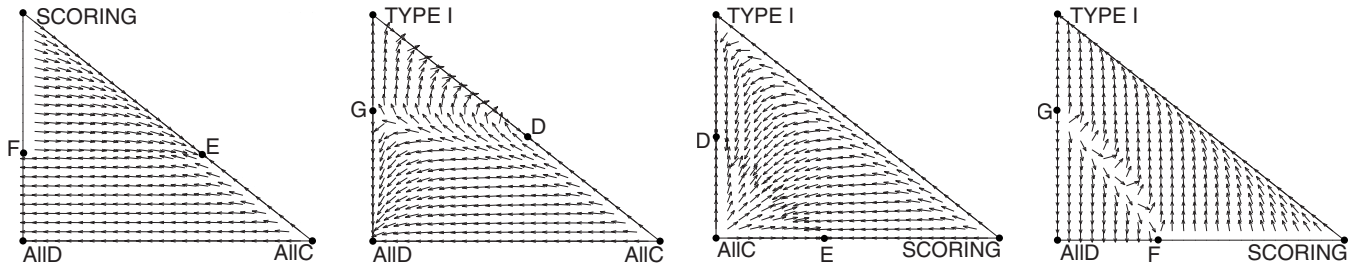


FIG. 6. Same as Fig. 5. Here, $q=0.1$.

It is clear that the structure on the All C-All D-SCORING surface is not affected by q since SCORING is a first-order assessment and only takes into account if help is actually given or not, and All C and All D players give help with a fixed probability. Therefore, the values of r_{ij} are independent of q for $i,j=1,2,3$, and so are the payoffs.

If TYPE I is included in the population, we find some qualitative differences from the case of public information. First, All C becomes able to invade the pure population consisting of TYPE I if q decreases. As a result, a new fixed point of the replicator dynamics is created in the segment between All C and TYPE I (point D). According to the simulations, the value of q at which the new fixed point occurs is between 0.8 and 0.85 for the parameter values chosen in this paper. This fixed point is stable within the segment.

Second, the fixed point in the segment between All C and SCORING is stabilized if q becomes small. For the perfect case, this mixed population is unstable since both TYPE I and TYPE II can invade the population [19]. But now TYPE I does not seem to be able to invade (see below for the analytical argument).

Overall, the merit of the second-order assessment rules TYPE I and TYPE II is reduced by private information. Nonetheless, it is possible that the population reaches a cooperative state since the polymorphic states including All C and TYPE I cannot be invaded by All D. On the other hand, SCORING becomes more stable and moreover plays an important role to achieve a cooperative state. The mixed state between All C and SCORING cannot be invaded by All D, TYPE I, or TYPE II.

The analytical approach supports these numerical outcomes. First, Eq. (3) for TYPE II yields the solution $r_{5i}=1/2$ regardless of i. This fact supports the finding obtained by the numerical simulations that the image matrix of TYPE II is random and does not depend on the types of observed players.

In Fig. 7, the replicator dynamics is displayed for the limit case of small q, which can be handled analytically. The star points between the points D and E and those between F and All D are the final states obtained by numerically integrating the replicator dynamics with various initial conditions. Depending on initial conditions, the state can evolve to two asymptotic regimes. One lies on the segment between All D and SCORING, with the frequency of defectors at least $1-c/\bar{\epsilon}b$. In this case, all defect. The other consists of cooperative polymorphic states where All C, SCORING, and TYPE I coexist. All the states connecting the two points D and E are stable, and on this segment cooperation is realized. The fixed point E is Lyapunov stable but not asymptotically stable since the whole segment E-F is a set of fixed points. The segment F-G also consists of fixed points, but they are unstable and not reachable.

V. DISCUSSION

The main findings of this paper are the following. First, the assessment rule of TYPE II does not work, except if all players have the same perfect information. Otherwise, the mismatch between the opinions of players spreads in the population, resulting in a random behavior: TYPE II behaves exactly like a mixed strategy playing the unconditional strategies All C and All D with the equal weight. Second, the pure population of TYPE I is not an evolutionarily stable strategy (ESS) any more: All C can invade if $q<1$. Nevertheless a cooperative state including TYPE I can be reached by the replicator dynamics. This cooperative state is necessarily polymorphic. Third, the fixed point in the segment between All C and SCORING becomes stable if the information q is sufficiently small. The second-order assessment rule TYPE I or All D cannot invade. This point, however, is not asymptotically stable. Fourth, the system is bistable for the case of small q: trajectories reach one of the cooperative polymorphic states consisting of All C, SCORING, and

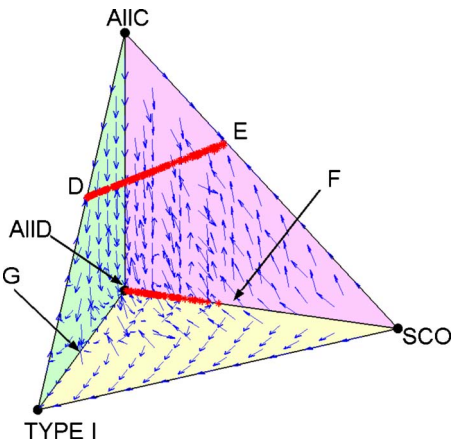


FIG. 7. (Color online) The vector field generated by replicator dynamics for the frequencies of All C, All D, SCORING, and TYPE I for the limit case of small q. TYPE II is not included. The vector at each point is normalized (except for the case where the vector vanishes), so that the direction is clearly recognized. The vector field of each surface is very similar to Fig. 6. Parameters: $c=1$, $b=3$, and $\epsilon=0.1$.

TYPE I, or one of the noncooperative states consisting of All D and SCORING, depending on where they start from.

In this paper, the imperfectness of information is based on both the direct and indirect observation models. Players in a population can obtain information about the interactions of others by direct observation or by rumor or gossip. The information given by a rumor is assumed to be always correct (for a model that includes incorrect rumors, see Nakamaru and Kawata [31]). In this sense, players "observe" the interactions indirectly in the latter case. The players assess the actions taken in the interactions according to their own assessment rules. Since any given interaction is not observed by all players (and different players may have observed the action of a given individual at different times), one and the same interaction can be differently interpreted by different TYPE I or TYPE II players.

In the literature, the imperfectness of information has been studied in various ways [13,32,33]. Nowak and Sigmund [13] considered the SCORING strategy under the assumption that each player either knows or does not know the reputation of another player and assumes a good reputation by default. They showed that the pure SCORING population cannot be invaded by All D players if the probability that a given player knows the reputation of others (denoted by q) is greater than the cost-benefit ratio: c/b. Brandt and Sigmund [32] found that cooperation can then be stably maintained if one allows q to grow with age. Mohtashemi and Mui [33] modified the model of Nowak and Sigmund [13] by assuming that the information spreads through social networks. They derived a condition for q which is similar to the inequality found by Nowak and Sigmund [13]. In those studies, the information is not perfect but still public. How this imperfectness affects the more complex higher-order assessments remains an open question.

There is another factor, not included in this paper, which can also lead to a mismatch between observers, namely, errors due to misperception [34]. If players misperceive whether help has been given or not, or if they are confused about the reputation of the recipient, the players can have different opinions even when they assess others in the same way. Introducing such errors seems particularly important for higher-order assessment rules. As we have seen in this paper, a mismatch of opinions affects second-order assessment rules to a greater extent than the more simple-minded first-order assessment. In fact, there is experimental evidence to support the view that second-order assessment can overtax human cognitive abilities [35].

The model investigated here has a considerable limitation. It concerns the assumption that players have only a very short memory: they determine their opinions of another individual according to their last observation of that individual's actions, and they do not take into account the earlier interactions. In real life, images are likely to be based on a longer memory. Moreover, the images are not binary: the moral world is not just black or white. It was not yet possible to overcome the complications raised by this factor.

According to Nowak and Sigmund [36], there are two motivations to pursue the investigation of indirect reciprocity. One concerns the evolution of human communities: how does cooperation work in villages and small-scale societies? [37–45]. The other motivation concerns the rapid growth of anonymous interactions on a global scale, which is made possible by the spread of communication networks: how can cheating be avoided in online trading? (see Bolton *et al.* [46] and Keser [47]). In both cases, simple robust methods for assessing others are needed. This paper has discussed the robustness of private assessment systems. Clearly this is no more than a step to investigate the competition between different moral systems under imperfect information.

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